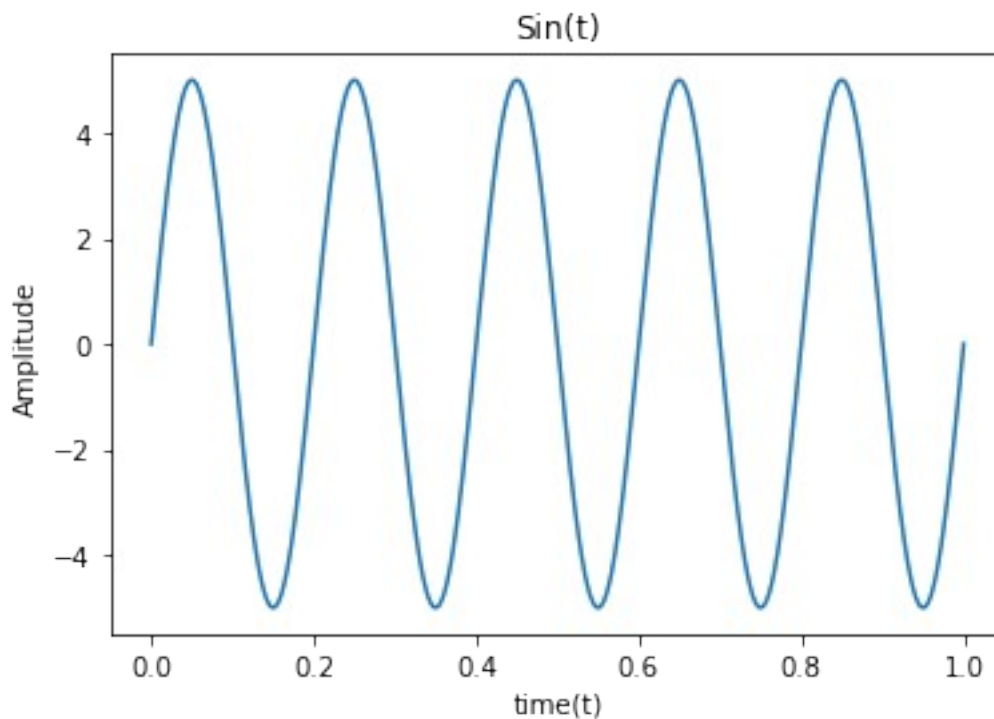


```
import matplotlib.pyplot as plt
import numpy as np

t = np.linspace(0,1,1000)
```

1.1 Generate and plot sinusoidal signal of 5V amplitude, 5Hz frequency and zero phase.

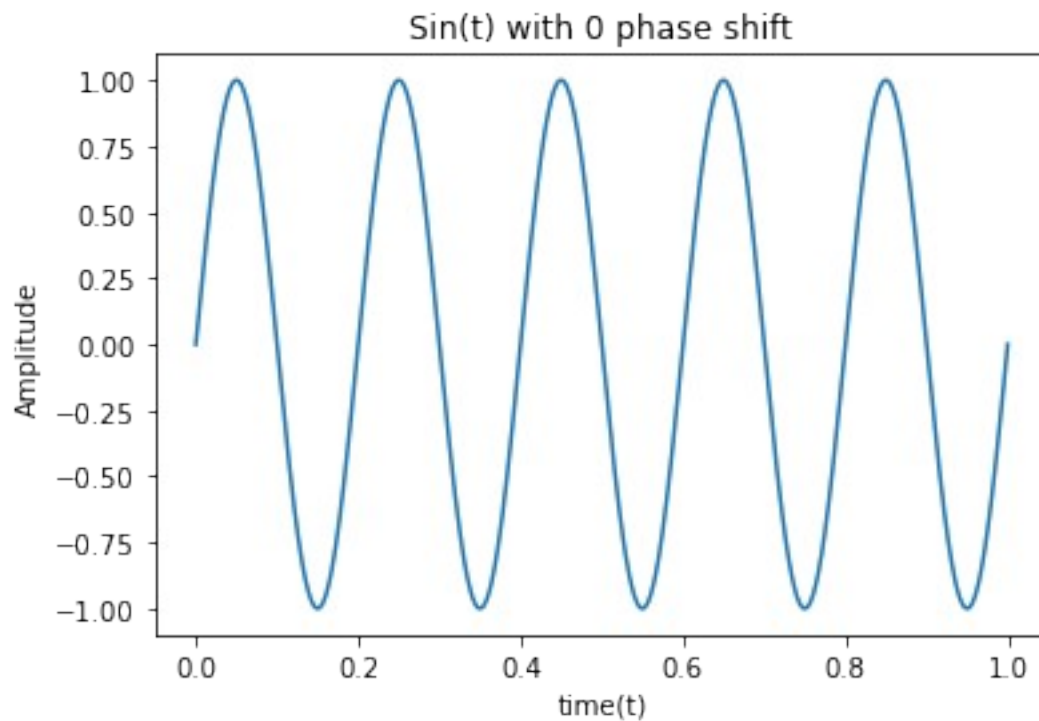
```
f = 5
plt.plot(t,5*np.sin(2*np.pi*f*t))
plt.xlabel("time(t)")
plt.ylabel("Amplitude")
plt.title("Sin(t)")
Text(0.5, 1.0, 'Sin(t)')
```



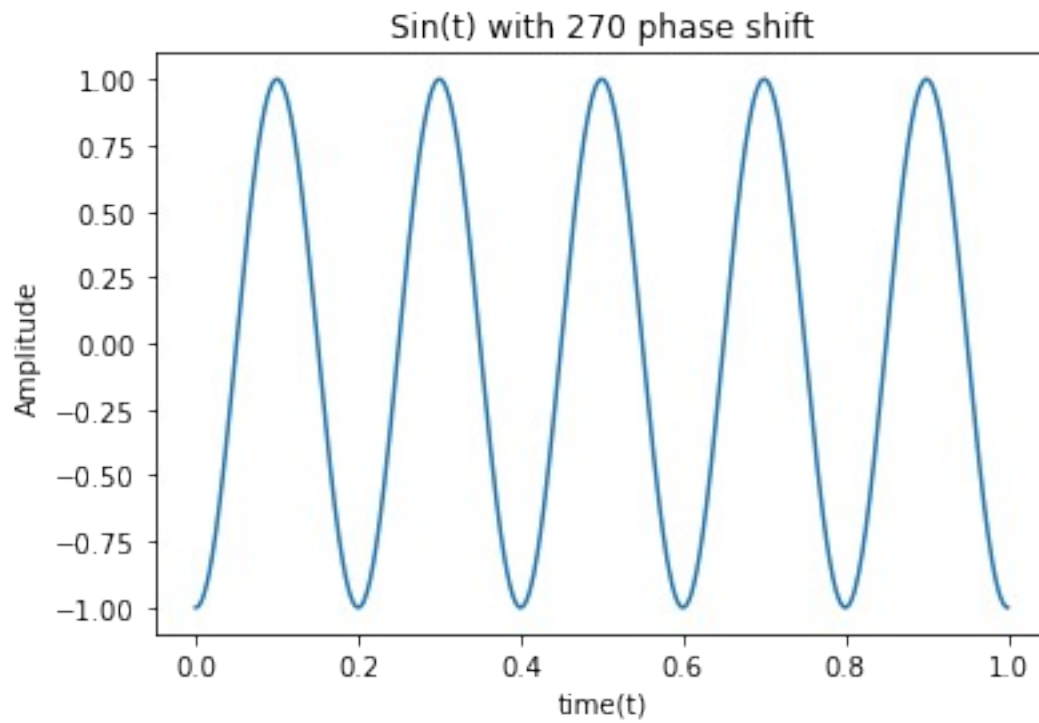
1.2 Generate and plot sinusoidal signals of 1V amplitude, 5Hz frequency with 0,90,180,270 phases in degrees.

```
a=1
f1=5
phi1 = 0
phi2 = (np.pi)/2
phi3 = np.pi
phi4 = (3*(np.pi))/2
```

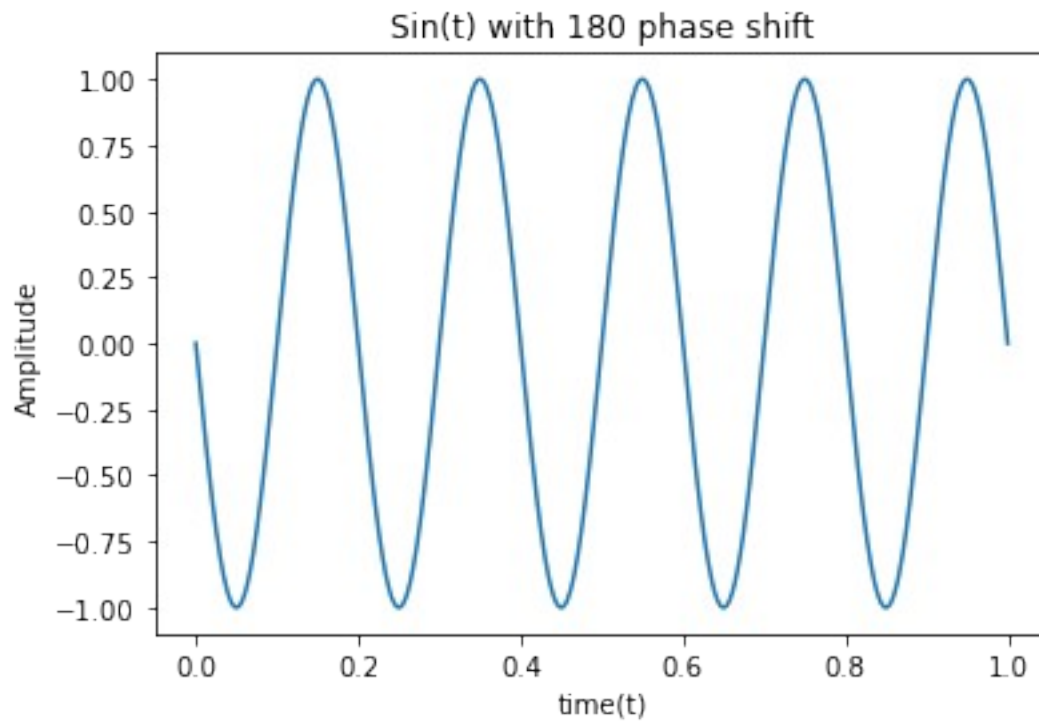
```
plt.plot(t,a*np.sin(2*np.pi*f1*t))
plt.xlabel("time(t)")
plt.ylabel("Amplitude")
plt.title("Sin(t) with 0 phase shift")
Text(0.5, 1.0, 'Sin(t) with 0 phase shift')
```



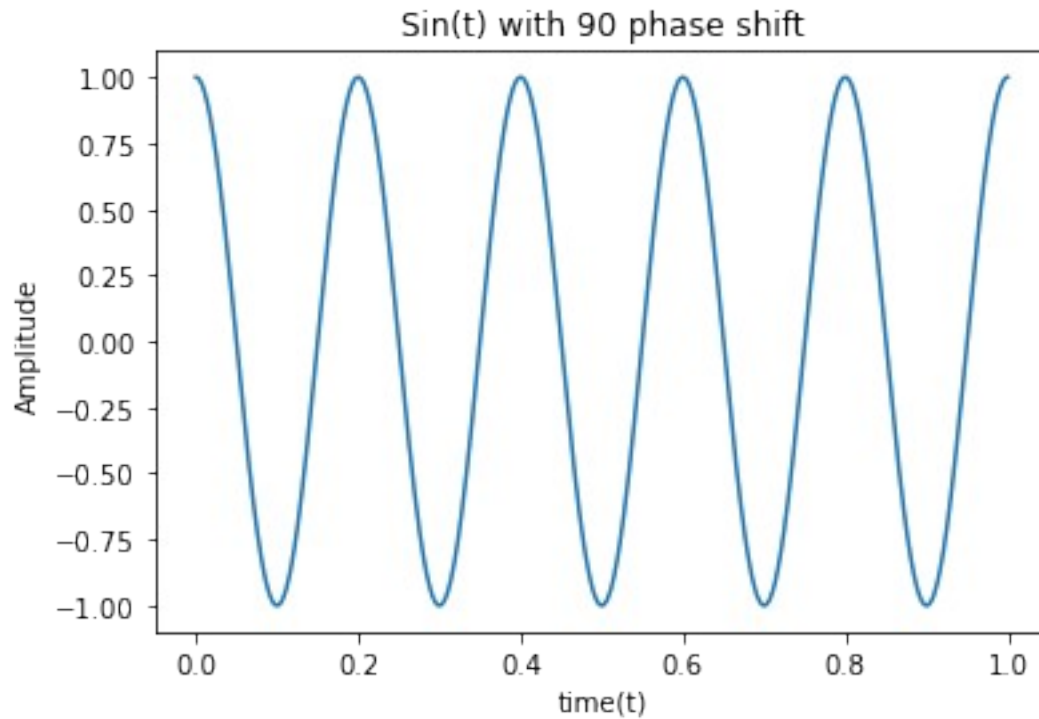
```
plt.plot(t,a*np.sin(2*np.pi*f1*t + phi4))
plt.xlabel("time(t)")
plt.ylabel("Amplitude")
plt.title("Sin(t) with 270 phase shift")
Text(0.5, 1.0, 'Sin(t) with 270 phase shift')
```



```
plt.plot(t,a*np.sin(2*np.pi*f1*t + phi3))  
plt.xlabel("time(t)")  
plt.ylabel("Amplitude")  
plt.title("Sin(t) with 180 phase shift")  
  
Text(0.5, 1.0, 'Sin(t) with 180 phase shift')
```



```
plt.plot(t,a*np.sin(2*np.pi*f1*t + phi2))
plt.xlabel("time(t)")
plt.ylabel("Amplitude")
plt.title("Sin(t) with 90 phase shift")
Text(0.5, 1.0, 'Sin(t) with 90 phase shift')
```

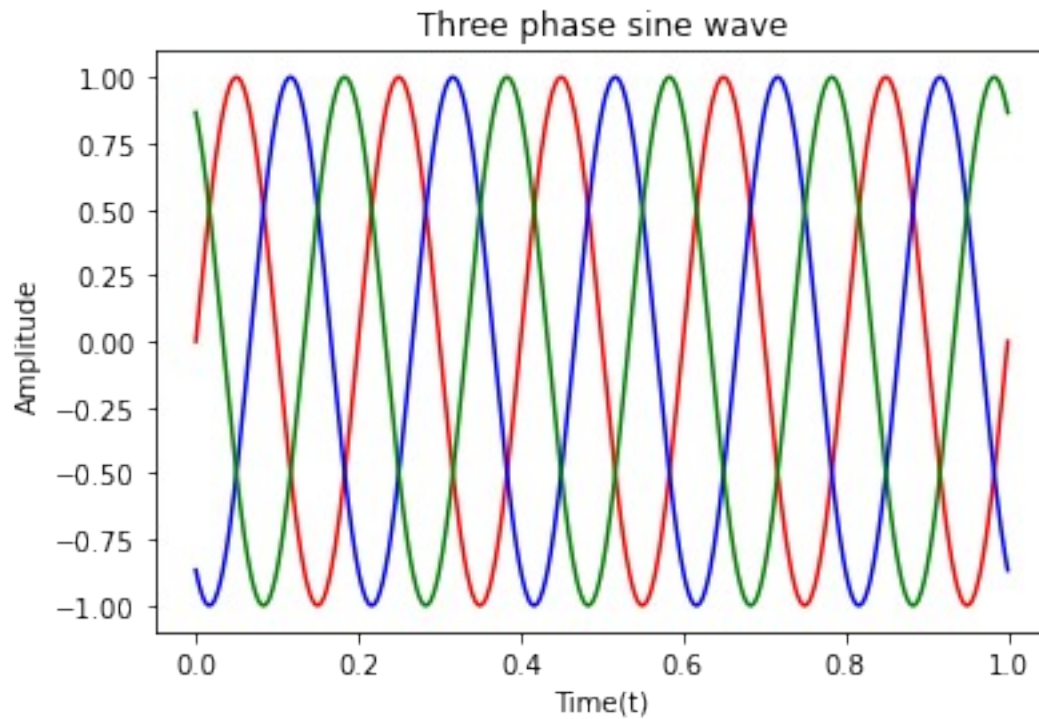


1.3 Generate and plot a three phase sine wave with 1V amplitude and 5Hz frequency.

```
y1 = np.sin(2*np.pi*5*t)
y2 = np.sin(2*np.pi*5*t-2 * np.pi / 3)
y3 = np.sin(2*np.pi*5*t-4 * np.pi / 3)

plt.plot(t,y1,'r')
plt.plot(t,y2,'b')
plt.plot(t,y3,'g')

plt.xlabel("Time(t)")
plt.ylabel("Amplitude")
plt.title("Three phase sine wave")
Text(0.5, 1.0, 'Three phase sine wave')
```

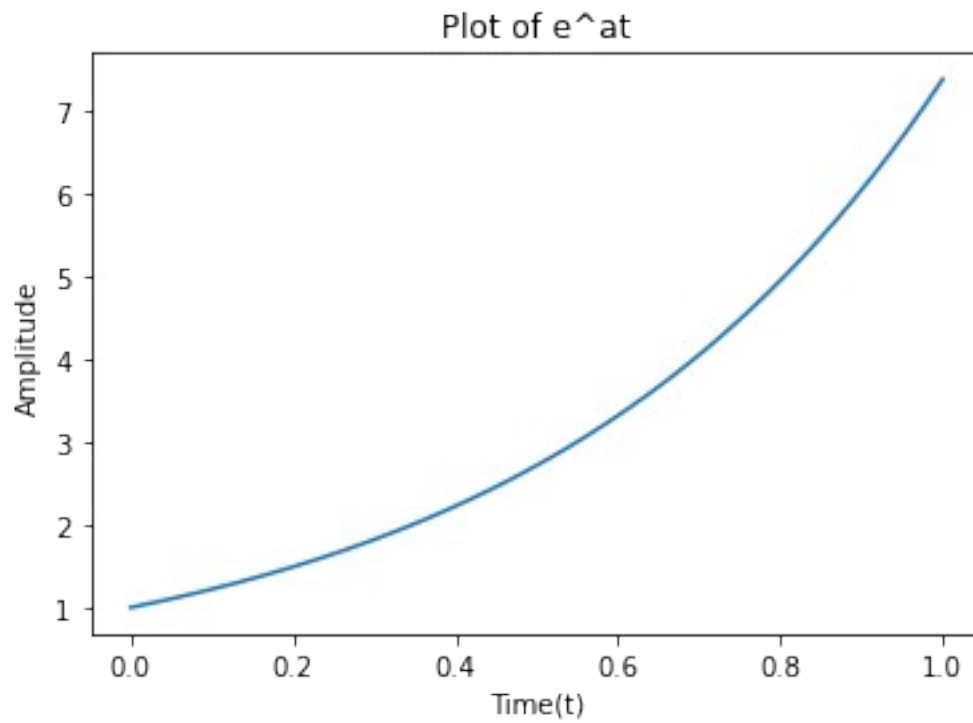


1.4 Generate and plot real exponential function signals of the form e^{at} with $a=2$ and -2 .

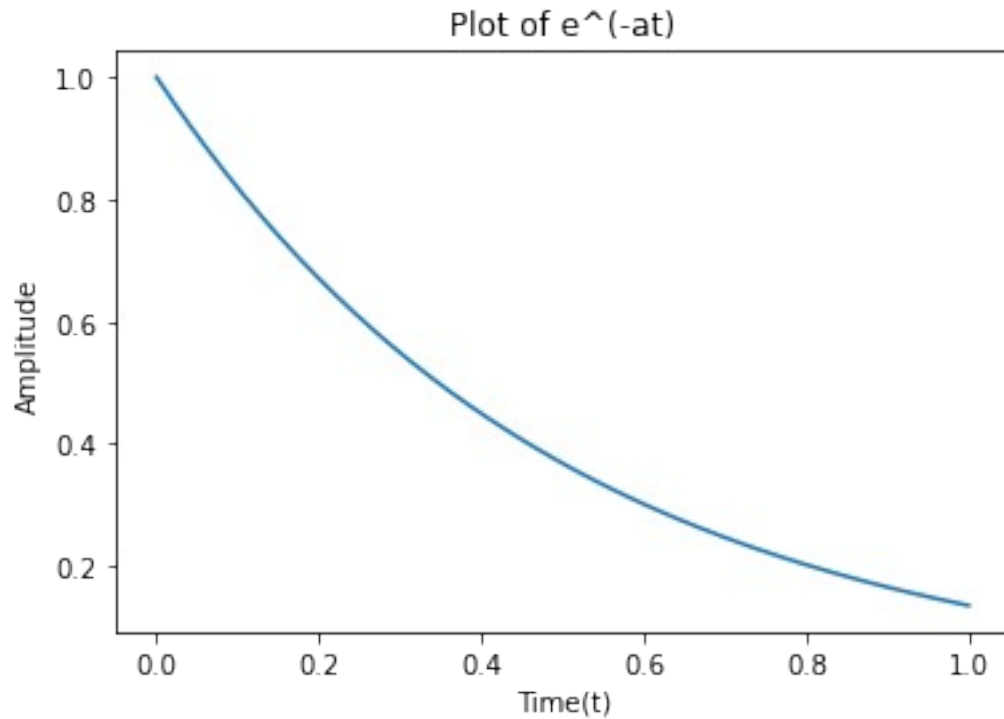
```
y4 = np.exp(2*t)
y5 = np.exp(-2*t)

plt.plot(t,y4)
plt.xlabel("Time(t)")
plt.ylabel("Amplitude")
plt.title("Plot of  $e^{at}$ ")

Text(0.5, 1.0, 'Plot of  $e^{at}$ ')
```



```
plt.plot(t,y5)
plt.xlabel("Time(t)")
plt.ylabel("Amplitude")
plt.title("Plot of  $e^{(-at)}$ ")
Text(0.5, 1.0, 'Plot of  $e^{(-at)}$ ')
```

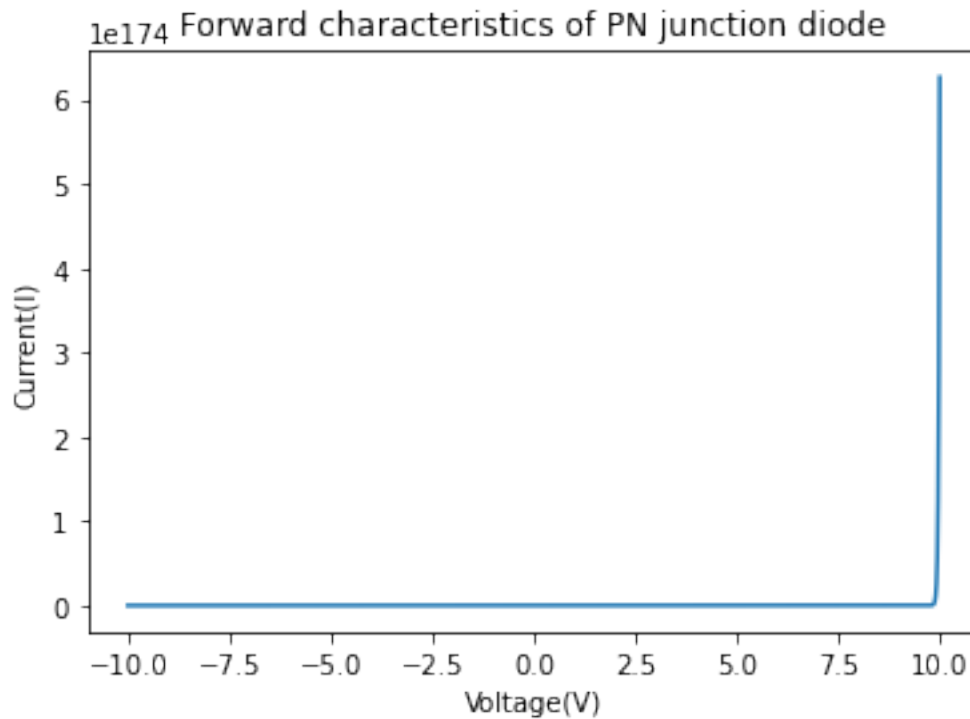


1.5 Generate and plot the forward characteristics of PN junction diode.

```
I_o = 12
V = np.linspace(-10,10,1000)
V_t = 0.025

y6 = I_o*(np.exp(V/V_t)-1)

plt.plot(V,y6)
plt.xlabel("Voltage(V)")
plt.ylabel("Current(I)")
plt.title("Forward characteristics of PN junction diode")
Text(0.5, 1.0, 'Forward characteristics of PN junction diode')
```

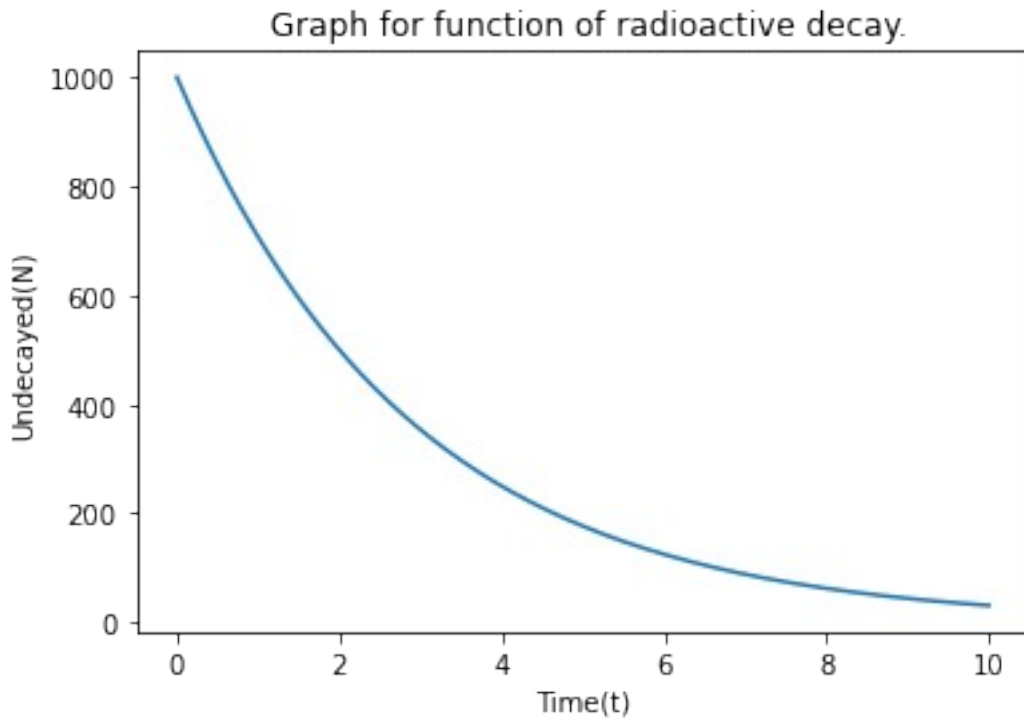


1.6 Generate and plot the function for radioactive decay.

```
N0 = 1000
half_life = 2.0
lam = np.log(2) / half_life

t2 = np.linspace(0, 10, 500)
N = N0 * np.exp(-lam * t2)

plt.plot(t2, N)
plt.xlabel("Time(t)")
plt.ylabel("Undecayed(N)")
plt.title(" Graph for function of radioactive decay.")
Text(0.5, 1.0, ' Graph for function of radioactive decay.')
```



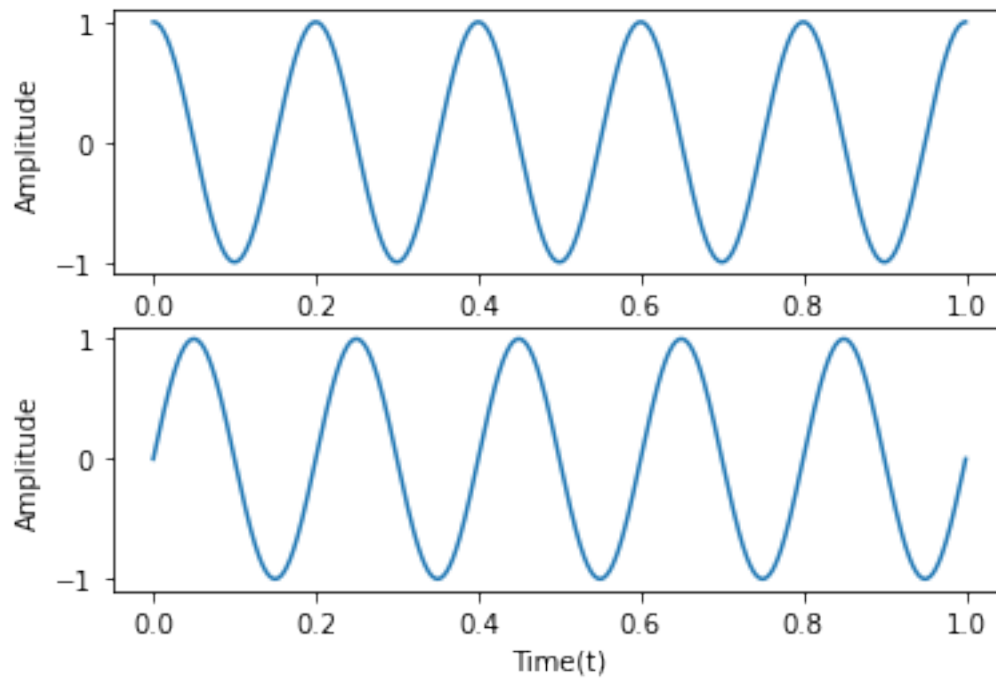
1.7 Generate and plot the complex exponential signals.

```
v1 = np.exp(1j*2*np.pi*5*t)
```

```
plt.subplot(2,1,1)  
plt.plot(t,np.real(v1))  
plt.xlabel("Time(t)")  
plt.ylabel("Amplitude")
```

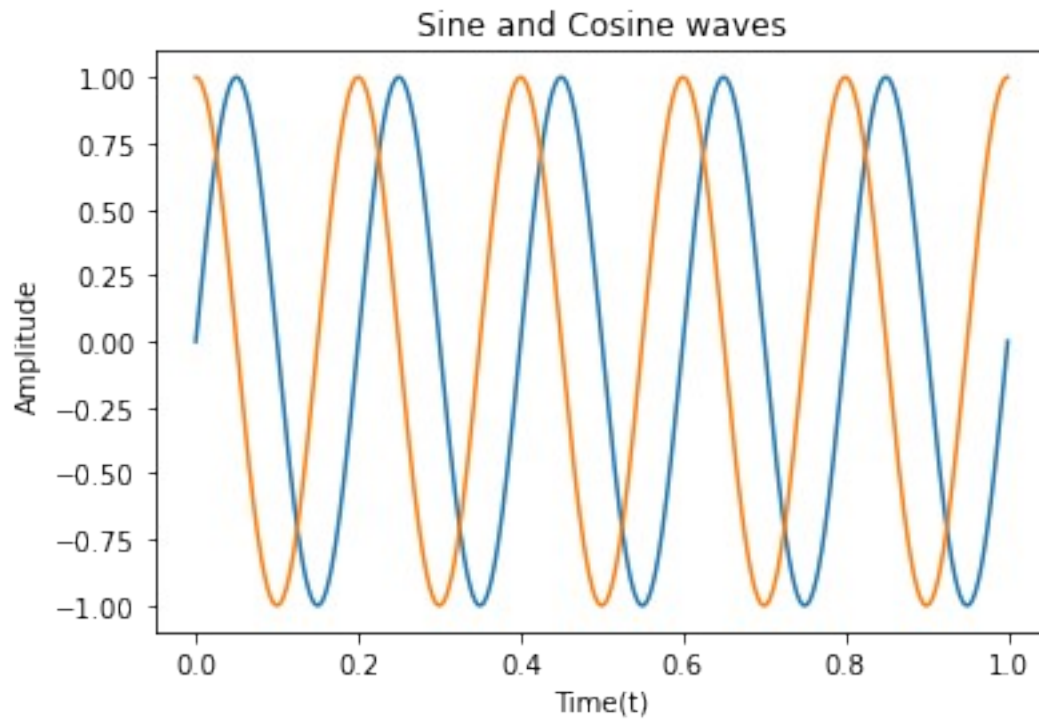
```
plt.subplot(2,1,2)  
plt.plot(t,np.imag(v1))  
plt.xlabel("Time(t)")  
plt.ylabel("Amplitude")
```

```
Text(0, 0.5, 'Amplitude')
```



1.8 Generate and plot the sine and cosine waves.

```
plt.plot(t,np.sin(2*np.pi*5*t))
plt.plot(t,np.cos(2*np.pi*5*t))
plt.xlabel("Time(t)")
plt.ylabel("Amplitude")
plt.title("Sine and Cosine waves")
Text(0.5, 1.0, 'Sine and Cosine waves')
```

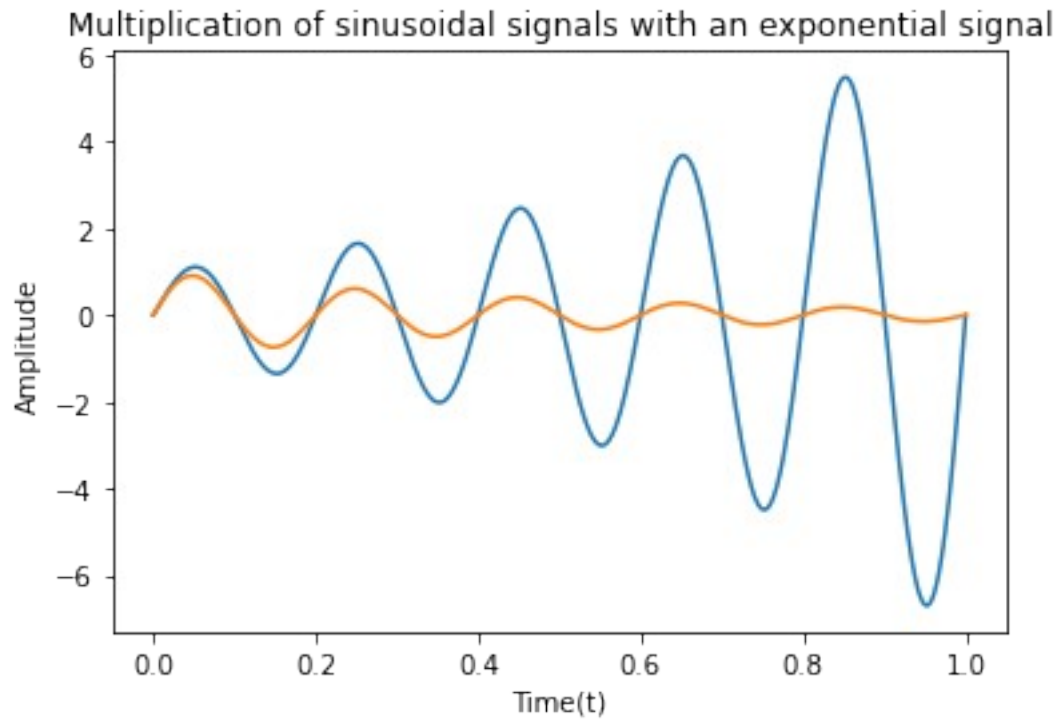


1.9 Generate and plot the outcome of multiplying sinusoidal signals with an exponential signal.

```
f1 = np.exp(2*t)
f2 = f1*np.sin(2*np.pi*5*t)
f3 = np.exp(-2*t)*np.sin(2*np.pi*5*t)

plt.plot(t,f2)
plt.plot(t,f3)
plt.xlabel("Time(t)")
plt.ylabel("Amplitude")
plt.title("Multiplication of sinusoidal signals with an exponential
signal")

Text(0.5, 1.0, 'Multiplication of sinusoidal signals with an
exponential signal')
```



1.10 Generate and compare the sine wave with variable t as t^2 and comment on the results.

```
p1 = np.sin(2*np.pi*5*t)
p2 = np.sin(2*np.pi*5*(t**2))

plt.plot(t,p1,'r')
plt.plot(t,p2,'b')
plt.xlabel("Time(t)")
plt.ylabel("Amplitude")
plt.title("sine wave with variable t as t^2")

Text(0.5, 1.0, 'sine wave with variable t as t^2')
```

